

will grow to be \$1,020, which is the amount of the cash payment provided in year ten. In other words, an investor would need to set aside \$836.76 today in order to have \$1,020 in ten years if the funds grew at a 2 percent annually compounded return.

If interest rates in the economy are 2.5 percent, then an investor would not be willing to pay \$1,000 for this bond that provides only 2 percent coupon payments. The investor would prefer a new bond that presumably is now offering a 2.5 percent coupon. To entice an investor to purchase the bond in this exhibit, the bond would have to be sold for a lower price. In a competitive and active market, bonds with the same maturity and risk characteristics must offer the same potential return for both parties to agree to a trade. In this case, the future cash flows are discounted at 2.5 percent, and the sum of these discounted cash flows (and potential selling price) is \$956.24.

Whoever owns the bond is entitled to the predefined cash flows of \$20 per year plus \$1,000 more in the final year. These cash flows do not change with interest rates. What changes is the selling price of the bond. For an investor who pays \$956.24 for the bond, the yield to maturity received by the investor on this smaller investment is 2.5 percent. Note that a 0.5 percent increase in interest rates reduced the selling price of the bond by 4.4 percent. If the investor sold the bond, the return received by the previous owner is defined in terms of any coupon payments received less the capital loss associated with the interest rate rise. This is why increasing interest rates lower the prices for existing bonds. The same phenomenon is also shown for an interest rate of 3 percent. In this case, the bond's price would have to be set at \$914.70 to adequately entice an investor. The 8.5 percent price reduction provides a yield to maturity of 3 percent to the new purchaser that then matches the overall higher interest rate in the economy.

Exhibit 7.1

Basic Pricing for a 10-Year Coupon Bond

Coupon Rate:	2%
Face Value:	\$1,000

Interest Rates:	2%	2.5%	3.0%
-----------------	----	------	------